

Thomas Gonzalez

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Skills

- Python | Java | JavaScript | C++ | React | Redux | Angular | Node.js | Django | Flask | Pandas | Scikit-learn | PyTorch | NumPy
- AWS | Kubernetes | Docker | Jenkins | CI/CD pipelines | IBM Cloud | REST APIs | Microservices | RDBMS | NoSQL (MongoDB, PostgreSQL) | HTML5 | CSS | Git | Oracle DB
- Agile | TDD | BDD | Unit Testing | Continuous Integration | Continuous Delivery | Cloud Computing | Frontend | Backend | Full-Stack Development

Experience

Software Engineer

General Dynamics - Electric Boat

Groton, CT, USA

10/2023 - Present

Hardware Validation Platform

- Architected and implemented a high-precision MySQL-backed data pipeline for real-time signal logging and hardware testing, capable of storing and exporting 28-day rolling datasets with millisecond-level accuracy.
- Designed scalable database schemas and RESTful Django APIs to facilitate data ingestion and retrieval for test simulations across multiple hardware platforms.
- Developed a Python-based frontend using tkinter integrated with backend APIs, allowing engineers to dynamically export filtered log data via pandas and numpy.
- Optimized system storage and retrieval operations, enabling lab-based hardware validation — accelerating project delivery timelines by over 150% compared to full in-field testing.

Battery Monitoring & Communication Software

- Contributed to full-stack development across a dual-language architecture (C++ backend, Java/JavaFX frontend) supporting UDP-based inter-process communication and real-time telemetry.
- Developed and extended checksum-protected messaging protocols over UDP; refactored threading and state-handling logic for higher runtime consistency.
- Increased test coverage from 60% to 98% by designing modular unit and integration tests (JUnit, GTest), significantly reducing regression incidents tested on Linux-based embedded systems.
- Led modernization of legacy systems by upgrading codebases to Java 23/JavaFX 23, improving stability, performance, and developer productivity.

Interactive Status Board

- Led full-stack development of a cross-ship coordination dashboard used to synchronize status updates and mission-critical visuals across the fleet.
- Maintained and enhanced a Java/JavaFX application running on Wildfly servers with EJB APIs and Oracle DB, designing new features with an emphasis on caching, security, and UI/UX improvements.
- Spearheaded quarterly Agile sprints in collaboration with naval operators; gathered feedback and deployed tailored builds across multiple classified environments.
- Upgraded backend infrastructure to leverage localized Wildfly caching, reducing compile time by 40% and improving reliability during low-bandwidth operations.

Education

Bachelor of Science

Binghamton University

Binghamton, NY, USA

08/2019 - 4/2023

- Major in Mathematical Physics

Projects

- **Sudoku Game:** Developed JavaFX Sudoku app with integrated ML for enhanced UX **(11/2023)**
- **Credit Card Recommendation App (In Development)** – Backend system integrating Plaid APIs with MySQL to provide real-time credit card recommendations based on Merchant Category Codes (MCCs). **(03/28/2025)**
- **Stock Price Predictor:** Developed and refined ML model for stock market forecasting. **(02/2024)**

Others

- **IBM Full Stack Certification:** Earned IBM Full Stack Cloud Developer certification, mastering end-to-end software development lifecycle (SDLC) methodologies and cloud deployments, with an intensive focus on programming languages including JavaScript and Python for building scalable, responsive applications.
- Implemented RESTful APIs, managed databases using MongoDB and PostgreSQL, and leveraged cloud services including IBM Cloud, enhancing application performance and user experience, showcasing advanced coding skills in JavaScript and Python.